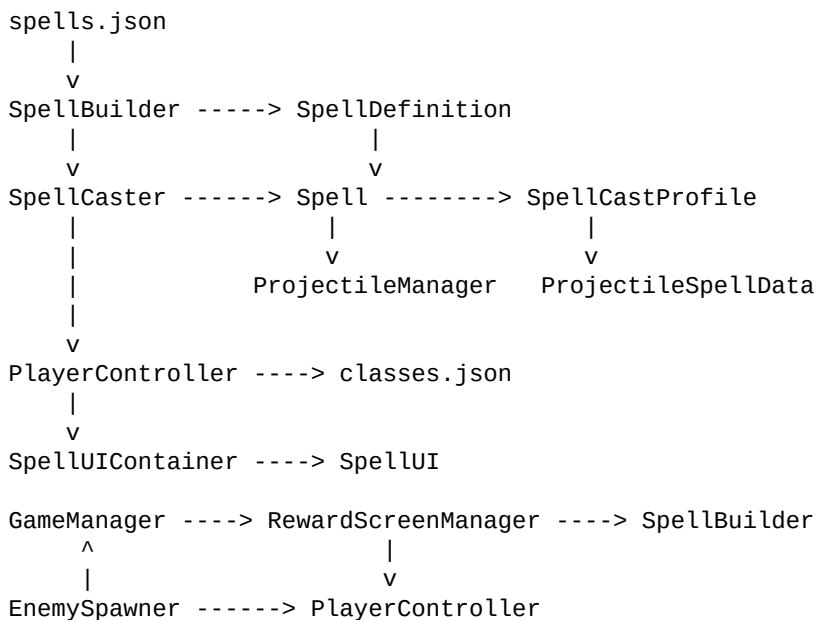


CMPM 121 Assignment 3 Report

Architecture Diagram



The main flow is data-driven. 'SpellBuilder' reads 'spells.json', stores every entry as a 'SpellDefinition', and separates base spells from modifier spells. 'SpellCaster' owns the player's mana, spell power, selected slot, and four spell slots. When the player casts, the selected 'Spell' builds a temporary 'SpellCastProfile', evaluates RPN expressions with 'wave' and 'power', applies modifiers, and asks 'ProjectileManager' to spawn the correct projectile behavior.

Wave rewards are coordinated by 'GameManager', 'RewardScreenManager', and 'EnemySpawner'. 'RewardScreenManager' creates and previews a pending random spell after a wave. The player can select an existing slot or drop a spell, and 'EnemySpawner.NextWave' equips the pending reward before the next wave starts.

Added And Changed Classes

'SpellDefinition': stores each JSON spell entry and exposes helpers for string fields, nested objects, and RPN integer/float evaluation. Methods: 'IsBaseSpell', 'GetString', 'GetObject', 'EvaluateInt', and 'EvaluateFloat'.

'ProjectileSpellData': stores projectile sprite, trajectory, speed, and lifetime, including JSON loading with fallback values. Methods: 'Copy' and 'FromJson'.

'SpellCastProfile': stores the final cast-time values after a base spell and its modifiers have been combined.

'Spell': now supports JSON-driven base spells and modifiers. It applies damage, mana, cooldown, speed, trajectory, delayed recasts, split volleys, spray projectiles, secondary projectiles, and on-hit projectile bursts. Methods: 'GetName', 'GetManaCost', 'GetDamage', 'GetDescription', 'GetCooldown', 'GetIcon', 'GetBaseId', 'GetModifierNames', 'IsReady', 'Cast', 'BuildProfile', 'ApplyModifiers', 'CastVolley', 'CastProjectiles', 'CreateProjectile', 'OnHit', 'CastSecondaryProjectiles', and

'CastOnHitProjectiles'.

'SpellBuilder': loads 'spells.json', builds the starting Arcane Bolt, builds random reward spells, and can build specific base/modifier combinations for testing. Methods: 'Build', 'BuildRandom', 'BuildReward', 'BuildWithModifiers', 'LoadDefinitions', 'LoadSpellData', and 'GetDefinition'.

'SpellCaster': now supports four spell slots, selected-slot casting, slot replacement, slot dropping, equipped-spell counts, and reward slot selection. Methods: 'ManaRegeneration', 'Cast', 'GetCurrentSpell', 'GetSpell', 'GetEquippedSpellCount', 'CanDropSpell', 'GetEquipSlotForNextSpell', 'SelectSpell', 'EquipSpell', 'EquipSpellAt', and 'DropSpell'.

'PlayerController': evaluates wave-scaled class stats from 'classes.json', preserves health and mana percentages while scaling, exposes spell selection, equipping, and dropping to the UI. Methods: 'StartLevel', 'EquipSpell', 'DropSpell', 'SelectSpell', 'EquipSpellAt', 'ApplyWaveStats', 'GetClassAttributes', 'LoadClassConfig', and 'EvaluateClassValue'.

'RewardScreenManager': previews the generated reward spell, shows damage/mana/cooldown, and tells the player which spell slot will receive or replace the reward. Methods: 'Start', 'Update', 'GetPlayer', 'EnsurePendingRewardSpell', 'GetMessage', 'GetRewardButtonText', 'GetRewardDescription', and 'GetRewardSlotDescription'.

'SpellUIContainer': keeps all four spell slots visible, refreshes each slot, highlights the selected slot, and forwards slot/drop commands. Methods: 'ConfigureSlots', 'ShowAllSlots', 'Refresh', 'SelectSlot0', 'SelectSlot1', 'SelectSlot2', 'SelectSlot3', 'SelectSlot', and 'DropSlot'.

'SpellUI': displays spell icon, mana cost, damage, cooldown fill, selection highlight, and enables the drop button only when dropping is valid. Methods: 'Start', 'Setup', 'SetSpell', 'Update', 'RefreshText', 'BindDropButton', 'UpdateDropButton', and 'DropSpell'.

'GameManager': stores the pending reward spell and reward wave so one reward is generated per completed wave. Methods: 'SetPendingSpellReward' and 'ClearPendingSpellReward'.

Spell Descriptions

Base spells:

- 'arcane_bolt': a straight projectile with medium damage.
- 'magic_missile': a low-damage homing projectile.
- 'arcane_blast': a medium projectile that creates secondary bolts on impact.
- 'arcane_spray': a cone of fast, short-lived low-damage projectiles.

Required modifiers:

- 'damage_amp': multiplies damage and mana cost.
- 'speed_amp': multiplies projectile speed.
- 'doubler': casts the modified spell again after a short delay.
- 'splitter': casts the modified spell in two slightly different directions.
- 'chaos': greatly increases damage and changes the projectile trajectory to spiraling.
- 'homing': changes the trajectory to homing, reduces damage, and adds mana cost.

Additional modifiers:

- 'mana_stream': lowers mana cost with a small cooldown penalty.
- 'overclocked': lowers cooldown and increases projectile speed, but raises mana cost.
- 'nova_burst': adds behavior by releasing extra arcane shards from the hit point.

Contributions

Todd Crandell implemented the JSON-driven spell pipeline, RPN stat evaluation, modifier application, random reward generation, wave-based player stat scaling, and the final reward-slot/drop polish. I learned that the spell system is easier to extend when a cast builds one temporary profile from data instead of putting every modifier into hard-coded branches.

ValerieVATS expanded the spell system from one equipped spell to a four-slot inventory and added the visible spell slot UI with selection and drop affordances. This work showed how important it is to keep UI as a view of gameplay state, while 'SpellCaster' remains the source of truth for what the player actually has equipped.